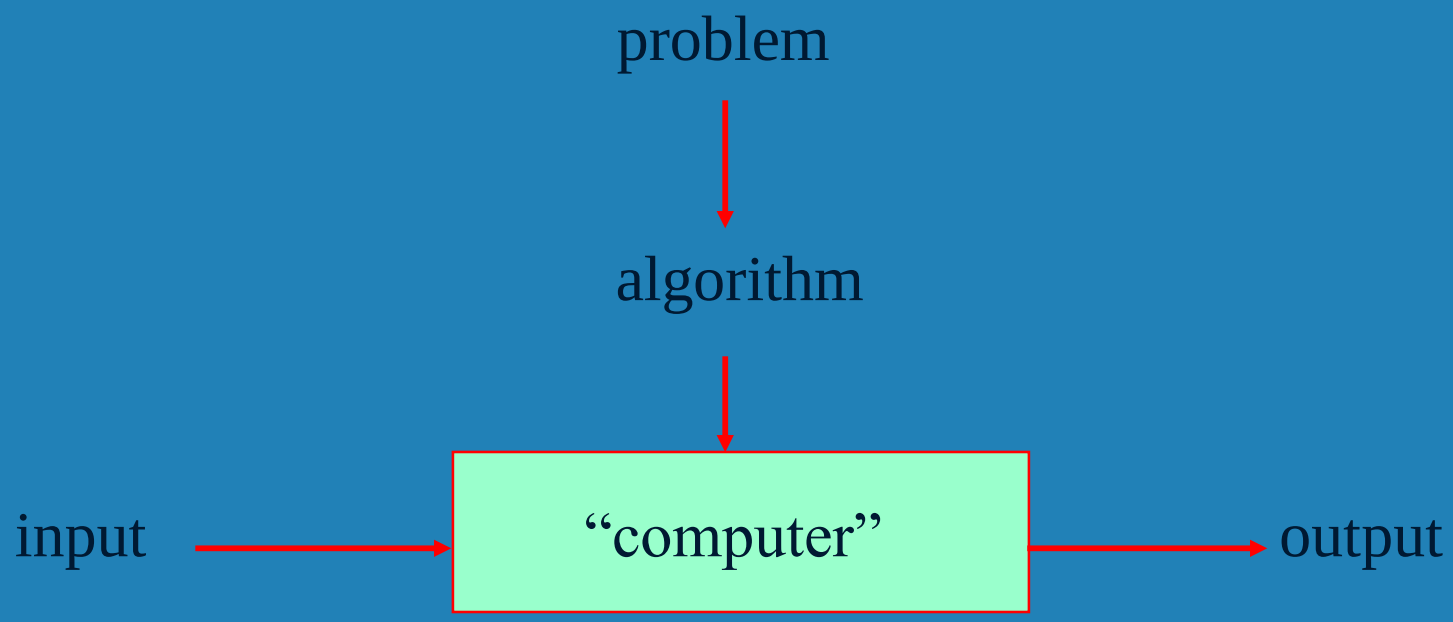
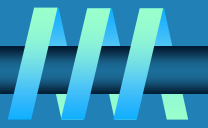
What is an algorithm?

An *algorithm* is a sequence of unambiguous instructions for solving a problem, i.e., for obtaining a required output for any legitimate input in a finite amount of time.



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Euclid's Algorithm



Problem: Find gcd(*m*,*n*), the greatest common divisor of two nonnegative, not both zero integers *m* and *n*

Examples: gcd(60,24) = 12, gcd(60,0) = 60, gcd(0,0) = ?

Euclid's algorithm is based on repeated application of equality

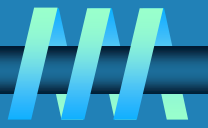
$$\gcd(m,n) = \gcd(n, m \bmod n)$$

until the second number becomes 0, which makes the problem trivial.

Example: gcd(60,24) = gcd(24,12) = gcd(12,0) = 12



Two descriptions of Euclid's algorithm



Step 1 If $n = 0$, return m and stop; otherwise go to Step 2

Step 2 Divide m by n and assign the value fo the remainder to r

Step 3 Assign the value of n to m and the value of r to n. Go to Step 1.

while $n \neq 0$ **do**

$r \leftarrow m \bmod n$

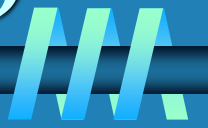
$m \leftarrow n$

$n \leftarrow r$

return m



Other methods for computing gcd(m,n)



Consecutive integer checking algorithm

Step 1 Assign the value of min{m,n} to t

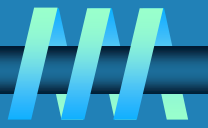
Step 2 Divide m by t. If the remainder is 0, go to Step 3; otherwise, go to Step 4

Step 3 Divide n by t. If the remainder is 0, return t and stop; otherwise, go to Step 4

Step 4 Decrease t by 1 and go to Step 2



Other methods for gcd(*m*,*n*) [cont.]



Middle-school procedure

Step 1 **Find the prime factorization of *m***

Step 2 **Find the prime factorization of *n***

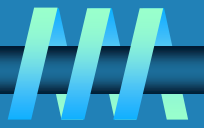
Step 3 **Find all the common prime factors**

Step 4 **Compute the product of all the common prime factors and return it as gcd(*m*,*n*)**

Is this an algorithm?



Sieve of Eratosthenes



Input: Integer $n \geq 2$

Output: List of primes less than or equal to n

for $p \leftarrow 2$ **to** n **do** $A[p] \leftarrow p$

for $p \leftarrow 2$ **to** $\lfloor \sqrt{n} \rfloor$ **do**

  if $A[p] \neq 0$ //p hasn't been previously eliminated from the list

    $j \leftarrow p * p$

    while $j \leq n$ **do**

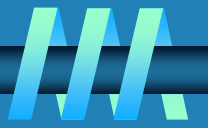
      $A[j] \leftarrow 0$ //mark element as eliminated

      $j \leftarrow j + p$

Example: 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

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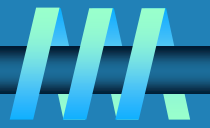
Why study algorithms?



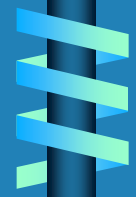
- Theoretical importance
  - the core of computer science
- Practical importance
  - A practitioner's toolkit of known algorithms
  - Framework for designing and analyzing algorithms for new problems



Two main issues related to algorithms



- How to design algorithms
- How to analyze algorithm efficiency

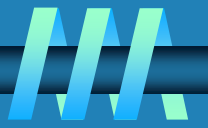


Algorithm design techniques/strategies

- Brute force
- Divide and conquer
- Decrease and conquer
- Transform and conquer
- Space and time tradeoffs
- Greedy approach
- Dynamic programming
- Iterative improvement
- Backtracking
- Branch and bound

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Analysis of algorithms

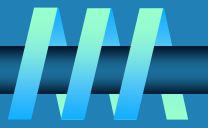


- How good is the algorithm?
  - time efficiency
  - space efficiency
- Does there exist a better algorithm?
  - lower bounds
  - optimality



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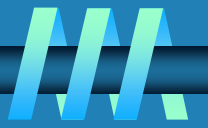
Important problem types



- sorting
- searching
- string processing
- graph problems
- combinatorial problems
- geometric problems
- numerical problems



Fundamental data structures



- list
  - array
  - linked list
  - string
- stack
- queue
- priority queue
- graph
- tree
- set and dictionary

